

# EXPERIMENTS

```
mysql> CREATE DATABASE college_exp19;
Query OK, 1 row affected (0.01 sec)

mysql> USE college_exp19;
Database changed
mysql>
mysql> CREATE TABLE student (
->     student_id INT,
->     student_name VARCHAR(50),
->     department VARCHAR(30),
->     marks INT
-> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> INSERT INTO student VALUES
-> (101, 'Arun', 'CSE', 85),
-> (102, 'Karthik', 'ECE', 78),
-> (103, 'Rahul', 'CSE', 92);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE TABLE student_backup (
->     student_id INT,
->     student_name VARCHAR(50),
->     department VARCHAR(30),
->     marks INT,
->     action VARCHAR(20)
-> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER after_student_insert
-> AFTER INSERT ON student
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO student_backup
->     VALUES (
->         NEW.student_id,
->         NEW.student_name,
->         NEW.department,
->         NEW.marks,
->         'INSERT'
->     );
-> END //
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> DELIMITER ;
mysql>
```

```
mysql> INSERT INTO student
-> VALUES (104, 'Vijay', 'IT', 88);
Query OK, 1 row affected (0.02 sec)
```

```
mysql>
mysql> SELECT * FROM student_backup;
+-----+-----+-----+-----+-----+
| student_id | student_name | department | marks | action |
+-----+-----+-----+-----+-----+
|          104 | Vijay        | IT         | 88    | INSERT |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE TRIGGER before_student_update
-> BEFORE UPDATE ON student
-> FOR EACH ROW
-> BEGIN
->     IF NEW.marks < 0 THEN
->         SET NEW.marks = 0;
->     END IF;
-> END //
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> UPDATE student
-> SET marks = -10
-> WHERE student_id = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql>
mysql> SELECT * FROM student;
+-----+-----+-----+-----+
| student_id | student_name | department | marks |
+-----+-----+-----+-----+
|          101 | Arun         | CSE        | 0     |
|          102 | Karthik      | ECE        | 78    |
|          103 | Rahul        | CSE        | 92    |
|          104 | Vijay        | IT         | 88    |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```